

## Clock gating

Low-power techniques come at the cost of speed, area and performance. Based on the application or system requirement, these need to be carefully adopted. In a system, a significant portion of dynamic power is consumed by the clock distribution network. Since clock buffers have the highest toggle rate in the system, these consume 50 per cent or even more dynamic power. Further, these have high drive strength to reduce clock delay.

In addition, flip-flops receiving the clocks can dissipate dynamic power even though these are not switching states. Designers can turn off the clocks for transistors or flip-flops when not functional. This helps in reducing a significant portion of dynamic power consumption, while preserving the state of transistors or flip-flops.

**Clock gating methods.** Following are the existing clock gating methods:

1. Gating based on inactive blocks/interfaces and protocol-defined states
2. Architectural clock gating implementation
3. Software program entry and exit
4. Hardware-driven and exit
5. Mixed hardware and software

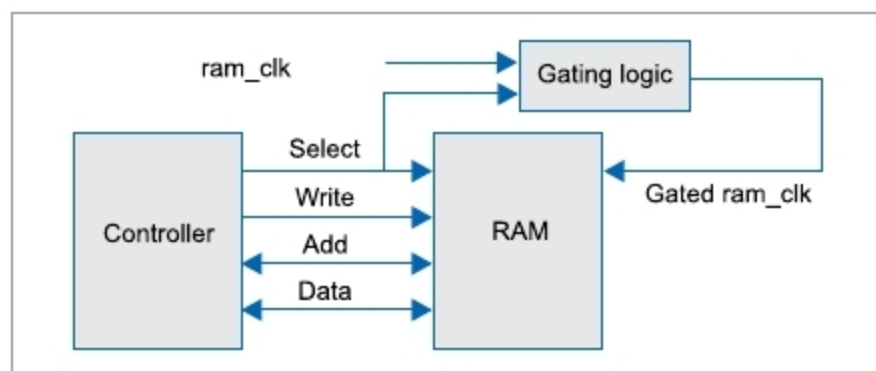


Fig. 1: Example of inactive blocks/interface gating

Consider a typical multi-bit flip-flop logic with enable (EN), as shown in Fig. 2. Updating of the flip-flop is done based on flip-flop enable; otherwise, the old value is retained.

The same circuit can be implemented with clock gating logic, as shown in Fig. 2. Clock gating reduces a significant amount of dynamic power consumed by the circuit. It also saves additional power and area by reducing the need for multiplexer logic at inputs of the flip-flops. This technique makes use of negative edge latch and AND gate to implement clock gating.

Today, most of the latest technology libraries include clock gating cells. Implementation tools can use these cells to perform auto-insertion operation of clock gating cells to reduce a considerable amount of dynamic power consumption.

## Architectural clock gating

Architectural clock gating in RTL is a configurable option in all versions of Cortex-M3.

In r1p1, it is controlled by several definitions and descriptions of licensees of the RTL.

```

define CLOCKGATE
define INTEGRATION_CLOCKGATE
  
```

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